Math 210B Lecture Notes

Lance Remigio

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1 Week 1

1.1 Basic Definition and Examples

Definition (Rings). (1) A **ring** is a set together with two binary operations $+$ and $\times$ (called addition and multiplication) satisfying the following axioms:

(i) $(R, +)$ is an Abelian group.

(ii) $\times$ is associative: $(a \times b) \times c = a \times (b \times c)$ for all $a, b, c \in \mathbb{R}$

(iii) The distribution law hold in R: For all $a, b, c \in R$

$$(a + b)\times = (a \times c) + (b \times c) \text{ and } a \times (b + c) = (a \times b) + (a \times c)$$

(2) The ring R is **commutative** if multiplication is commutative.

(3) The ring R is said to have an **identity** (or contain 1) if there is an element $1 \in R$ with

$$a \times 1 = 1 \times a = a \quad \forall a \in R.$$

Definition (Division Ring, Field). A ring R with identity $1 \neq 0$ is called a **division ring** if any nonzero element $a \in R$ has a multiplicative inverse; that is,

$$\exists b \in R \text{ such that } ab = ba = 1.$$

A commutative division ring is called a **field**.

Proposition. Let R be a ring. Then

(1) $0a = a0 = 0$ for all $a \in R$.

(2) $(-a)b = a(-b) = -(ab)$ for all $a, b \in R$ where $-a$ is the additive inverse of a

(3) $(-a)(-b) = ab$ for all $a, b \in R$.

(4) If R has identity 1, then the identity is unique and $-a = (-1)a$.

Definition (Zero Divisors, Units). (1) We call $a \neq 0$ in R a **zero divisor** if there exists $b \neq 0$ in R such that either $ab = 0$ or $ba = 0$.

(2) Assume $1 \neq 0 \in R$. We say that $u \in R$ is a **unit** if there exists $v \in R$ such that

$$uv = vu = 1.$$

We denote the set of units of R as $R^\times$. (group of units of R)

- (2) tells us that the set of units form a group under multiplication.

- Zero divisors can never be units.

This result tells us that fields do not have zero divisors.

Definition (Integral Domain). A commutative ring with identity $1 \neq 0$ is called **integral domain** if it has no zero divisors.

Rings that are commutative gives rise to the cancellation property.

Proposition. Let R be a ring and let $a, b, c \in R$ such that a is not a zero divisor. Then the following are true: If $ab = ac$, then either $a = 0$ or $b = c$. In particular, if a, b, c are any elements in an integral domain and $ab = ac$, then either $a = 0$ or $b = c$.

Corollary. Any finite integral domain is a field.

Another perspective given by the cancellation law is that every nonzero element of a commutative ring that is not a zero divisor has a multiplicative inverse in a larger ring. This naturally leads us to the discussion of subrings.

Definition (Subring). A **subring** of the ring R is a subgroup of R that is closed under multiplication.

To show that a subset of a ring is a subring, it is enough to show that S is nonempty and S is closed under multiplication and subtraction (much like the subgroup criterion in 210A).

1.2 Examples of Rings

Definition (Polynomials). (1) Let x be an indeterminate. The formal sum

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \qquad (n \geq 0)$$

with each $a_i \in R$ $(1 \leq i \leq n)$ is called a **Polynomial** in x with **coefficients** $a_i \in R$.

(2) If $a_n \neq 0$, then the polynomial is said to have **degree** n, and $a_n x^n$ is called the **leading term**, and a_n is called the **leading coefficient** (Note: the leading coefficient of the zero polynomial is taken to be zero).

(3) We say that the polynomial is **monic** if $a_n = 1$.

Definition (Polynomial Rings). The set of polynomials in the variable x with coefficients in R is called a **polynomial ring** and is denoted by $R[x]$.

Addition and multiplication defined on $R[x]$ are the operations familiar from elementary algebra.

- Addition in $R[x]$ is defined component-wise; that is,

$$\begin{aligned}&(a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0) + (b_n x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0)\\ &= (a_n + b_n)x^n + (a_{n-1} + b_{n-1})x^{n-1} + \cdots + (a_1 + b_1)x + (a_0 + b_0).\end{aligned}$$

- Multiplication is defined in the following way:

$$\begin{aligned}&(a_0 + a_1 x + a_2 x^2 + \cdots) \times (b_0 + b_1 x + b_2 x^2 + \cdots)\\ &= a_0 b_0 + (a_0 b_1 + a_1 b_0)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 + \cdots\end{aligned}$$

In general, the coefficients of x^k in the product will be $\displaystyle\sum_{i=0}^{k} a_i b_{k-i}$.

Example (Examples of Polynomial Rings).

- $\mathbb{Z}[x]$ is the polynomial ring with integer coefficients;
- $\mathbb{Q}[x]$ is the polynomial ring with rational coefficients;
- $\mathbb{Z}/3\mathbb{Z}[x]$ is the polynomial ring with coefficients in $\mathbb{Z}/3\mathbb{Z}$.

Here is an example computation: If $p(x) = x^2 + \overline{2}x + \overline{1}$ and $q(x) = x^3 + x + \overline{2}$, then

$$\begin{aligned} p(x) + q(x) &= x^3 + x^2 + (\overline{2} + \overline{1})x + (\overline{1} + \overline{2}) \\ &= x^3 + x^2 + \overline{3}x + \overline{3} \\ &= x^3 + x^2. \end{aligned}$$

Proposition (Proposition 4 Dummit and Foote). Let R be an integral domain and let $p(x), q(x)$ be nonzero elements of $R[x]$. Then

(1) The degree of $p(x)q(x) =$ the degree of $p(x)$ + the degree of $q(x)$.

(2) The units of $R[x]$ are just units of R.

(3) $R[x]$ is an integral domain.

Proof. (1) Let $p(x)$ and $q(x)$ be polynomials in $R[x]$ with leading terms $a_n x^n$ and $b_m x^m$, respectively. Then the leading term of $p(x)q(x)$ is $a_n b_n x^{n+m}$ with $a_n b_m \neq 0$. This tells us that $\deg[p(x)q(x)] = \deg[p(x)] + \deg[q(x)]$.

(2) If $p(x)$ is a unit in $R[x]$, then there exists a polynomial $q(x)$ such that $p(x)q(x) = 1$. This tells us that $\deg\, p(x) + \deg\, q(x) = 0$, so both $p(x)$ and $q(x)$ are elements of R, and so are units in R since their product is 1.

(3) If R has no zero divisors, then neither does $R[x]$ (because $R \subseteq R[x]$).

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- If R does have zero divisors, then $R[x]$ also has them since $R \subseteq R[x]$.
- If $f(x)$ is a zero divisor in $R[x]$, then $cf(x) = 0$ for some nonzero $c \in R$.
- If S is a subring of R, then $S[x]$ is a subring of $R[x]$.

2 Week 2

2.1 Ring Homomorphisms and Quotient Rings

Definition (Ring Homomorphisms, Ring Isomorphisms, and Kernel). Let R and S be rings.

(1) A **ring homomorphism** is a map $\varphi : R \to S$ satisfying

  (i) $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$. (This is regarded as the group homomorphism for additive groups)

  (ii) $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$. That is, φ is multiplicative.

(2) The **kernel** of the ring homomorphism φ, denoted by $\ker(\varphi)$ is

$$\ker(\varphi) = \{x \in R : \varphi(x) = 0_s\}$$

where 0_s is additive identity.

(3) A bijective ring homomorphism is called an **isomorphism**.

Example. Define $\varphi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ by

$$\varphi(x) = \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd.} \end{cases}$$

It is relatively straightforward to check that this satisfies both (i) and (ii) in the definition above.

Proposition (Proposition 5 from Dummit and Foote)**.** Let R and S be rings and let $\varphi : R \to S$ be a homomorphism.

(1) The **image** of φ is a subring of S.

(2) The **kernel** of φ is a subring of R.

Furthermore, if $\alpha \in \ker(\varphi)$ and $r \in R$, then $r\alpha \in \ker(\varphi)$ and $\alpha r \in \ker(\varphi)$ (we call this the absorptive property of $\ker(\varphi)$).

2.2 Ideals and Quotient Rings

Note that since R is a ring and its additive structure is forced to be abelian by the distributive and associative ring axioms, it follows that every subgroup of R is a normal subgroup. That is, when adding two cosets together (with I as our additive normal subgroup of R), we get that for any $x+I, y+I \in R/I$

$$(x+I) + (y+I) = (x+y) + I.$$

This tells us that R/I has a group structure. However, is it possible to make R/I into a ring? Of course! The extra structure that we need here is for I to be an ideal.

Definition (Ideals)**.** Let R be a ring, let I be a subset of R and let $r \in R$.

(1) $rI = \{ra : a \in I\}$ and $Ir = \{ar : a \in I\}$

(2) A subset I is a **left ideal** of R if

  (i) I is a subring of R

  (ii) I is closed under left-multiplication by elements from R i.e $rI \subseteq I$ for all $r \in R$. Similarly, I is a **right ideal** if (i) holds and I is closed under right-multiplication by elements of R i.e $Ir \subseteq I$.

  (3) A subset I that is both a left and right ideal is called an **ideal** (more specifically, a two-side ideal) of R.

Example. The subring defined by $I = \{p(x) \in \mathbb{Z}[x] : p(x) = 0 \text{ or } \deg(p(x)) \geq 2\}$ is an ideal in $\mathbb{Z}[x]$.

It is clear that I is additive subgroup of $\mathbb{Z}[x]$. Let $f(x)$ be a polynomial in $\mathbb{Z}[x]$ and $g(x) \in I$. Then we have

$$f(x) = \sum_{i=0}^{n} a_i x^i \text{ and } \sum_{i=1}^{n} b_i x^i$$

where $b_0 = 0$ and $b_1 = 0$. Note that the constant term of $f(x)q(x)$ is $a_0 b_0 = a_0 0 = 0$ and the linear term is $a_1 b_0 + a_0 b_1 = a_1 0 + a_0 0 = 0$.

Proposition (Proposition 5 from Dummit and Foote)**.** Coset multiplication in R/I is well-defined if and only if I is an ideal.

Proposition (Proposition 6 from Dummit and Foote)**.** Let R be a ring and let I be an ideal of R. Then the (additive) quotient group R/I is a ring under the binary operations:

$$(r+I)+(s+I)=(r+s)+I \text{ and } (r+I)(s+I)=(rs)+I \text{ for all } r,s\in R.$$

If I is any subgroup of R satisfying the above operations, then I is an ideal of R.

Definition (Quotient Ring)**.** When I is an ideal of R, the ring R/I with the operations defined in the previous proposition is called the **quotient ring of** R **by** I .

Theorem (First Isomorphism Theorem)**.** (1) If $\varphi : R \to S$ is a homomorphism of rings, then the kernel of φ is an ideal of R, the image of φ is a subring of S and

$$R/\ker(\varphi) \cong \varphi(R).$$

(2) If I is any ideal of R, then the map $R \to R/I$ defined by $r \mapsto r+I$ is a surjective ring homomorphism with kernel I. In this case,

$$I \text{ is an ideal if and only if } I \text{ is the kernel of some ring homomorphism.}$$

Example. (1) Let R be a ring. The trivial ideals of R are R itself and $\{0\}$ (the zero ideal).

(2) Last semester (when studying cyclic groups) we proved that the only subgroups of $\mathbb{Z}$ were of the form $n\mathbb{Z}$ where $n \geq 0$. This tells us that al the ideals of $\mathbb{Z}$ must be of the same form. It is a straightforward exercise to show that $n\mathbb{Z}$ ($n \geq 0$) is an ideal of $\mathbb{Z}$.

Example. Let A be a ring and X be a nonempty set and let R be a ring of all functions from X to A. For each c, we can define a mapping $E_c : R \to A$ by $E_c(f) = f(c)$ (This is called the evaluation map). It follows that for any $f, g \in R$, we have

$$E_c((f+g)(x)) = (f+g)(c) = f(c)+g(c) = E_c(f(x)) + E_c(g(x))$$

and

$$E_c\Big((fg)(x)\Big) = (fg)(c) = f(c)g(c) = E_c(f(x))E_c(g(x)).$$

Also,

$$\begin{aligned}\ker(E_c) &= \{f \in R : E_c(f) = 0\}\\ &= \{f \in R : f(c) = 0\}.\end{aligned}$$

Using the first isomorphism theorem, we obtain

$$R/\ker(E_c) \cong E_c(R). \qquad (*)$$

Let $a \in A$. Notice that $h(x) = a$ (the constant function) gives that $E_c(h(x)) = h(c) = a$. Thus, we can say that E_c is surjective and that (*) can be re-written to

$$R/\ker(E_c) \cong A.$$

Theorem (The Last Three Isomorphism Theorems)**.** (1) **(The Second Isomorphism Theorem)** Let A be a subring and let B be an ideal of R. Then $A+B$ is a subring of R, $A \cap B$ is an ideal of A, and

$$(A+B)/B \cong A/(A \cap B).$$

(2) **(The Third Isomorphism Theorem for Rings)** Let I and J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and

$$(R/I)/(J/I) \cong R/J.$$

(3) **(The Fourth or Lattice Isomorphism Theorem for Rings)** Let I be an ideal of R. There exists a one-to-one correspondence between the set of subrings A of R that contain I and the set of subrings of R/I. That is,

$$A \text{ is an ideal in } R \text{ that contains } I \text{ if and only if } A/I \text{ is an ideal of } R/I.$$

Definition (Sum, Product, Powers of Ideals). (1) Define the **sum** of I and J by $I + J = \{a + b : a \in I, b \in J\}$.

(2) Define the **product** of I and J, denoted by IJ, to be the set of all finite sums of elements of the form ab with $a \in I$ and $b \in J$.

(3) For all $n \geq 1$, define the **nth power of** I, denoted by I^n, to be the set of all finite sums of elements of the form $a_1 a_2 \ldots a_n$ with $a_i \in I$ for all i. Equivalently, I^n is defined inductively by defining $I^1 = I$ and $I^n = II^{n-1}$ for $n = 2, 3, \ldots$.

- $I + J$ is the smallest ideal containing both I and J.
- IJ is contained in $I \cap J$.

Corollary. If R is a field, then any nonzero ring homomorphism from R into another ring S in an injection.

Definition (Maximal Ideal). An ideal M is a ring is called a **maximal ideal** if it is proper (that is, $M \neq S$) and the only ideals which contain M are S and itself; that is, for any ideal $K \leq R$, if $M \leq K \leq R$, then $M = K$ or $K = R$.

Proposition. In a ring with identity, every proper ideal is contained in a maximal ideal.

Proposition. Assume R is commutative. The ideal M is a maximal ideal if and only if R/M is a field.

Example. What are the maximal ideals in $\mathbb{Z}$? We know all subrings of $\mathbb{Z}$ are of the form in $n\mathbb{Z}$ for $n \geq 0$. It is straightforward to show that $n\mathbb{Z}$ is an ideal. The maximal ideals are those ideals such that $\mathbb{Z}/n\mathbb{Z}$ is a field. We know that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime; that is, $n\mathbb{Z}$ is a maximal ideal if and only if p is prime.

Definition (Prime Ideals). Assume R is commutative ring. An ideal P is a prime ideal if $P \neq R$ and if $ab \in P$, then $a \in P$ or $b \in P$.

Remark (Motivation for Prime Ideals). Look at $\mathbb{Z}$. Suppose $n\mathbb{Z}$ ($n \geq 0$) is a prime ideal of $\mathbb{Z}$. Then if $ab \in n\mathbb{Z}$, we have that $a \in n\mathbb{Z}$ or $b \in n\mathbb{Z}$. Hence, if we say that $a \in (n)$, then this is just $a = nx$ for some $x \in \mathbb{Z}$; that is, we are saying that $n \mid a$.

Equivalently, we can say that if $n \mid ab$, then either $p \mid a$ or $p \mid b$. Prime ideals reflect this kind of a behavior.

Proposition. Assume R is commutative. The ideal P is a prime ideal in R if and only if R/P is an integral domain.

Corollary. Every maximal ideal is a prime ideal when our ring is commutative.

3 Week 3

3.1 More on Ideals

Proposition. Let I and J be ideals of a ring R. Define $I + J$ as $\{a + b : a \in I \text{ and } b \in J\}$. Then $I + J$ is an ideal of R and it is the smallest ideal that contains both I and J.

Proof. Note that I and J are subrings of R and so they are both normal additive subgroups of R. Hence, $I + J$ is an additive subgroup of R.

Let $r(x + y) \in r(I + J)$ where $x \in I$ and $y \in J$. Then

$$r(x + y) = rx + ry.$$

But note that $rx \in I$ and $ry \in J$. Thus, $r(x + y) \in I + J$. Similarly, if $(x + y)r \in (I + J)r$, then

$$(x + y)r = xr + yr$$

where $xr \in I$ and $yr \in J$ since I and J are ideals of R. Thus, $(x + y)r \in I + J$. This tells us that $I + J$ is an ideal of R.

Notice that for containment, it follows that since $I + J$ is closed under addition, we have for every $x \in I$, $x = x + 0_R$ implies that $x \in I + J$ and likewise, if $x \in J$, then $x = 0_R + x$ implies that $x \in I + J$. Thus, both ideals are contained in $I + J$. To see that $I + J$ is the smallest ideal containing I and J, take any arbitrary ideal K in R such that K contains both I and J. Since K is closed under addition, it follows that $I + J \subseteq K$ and so $I + J$ ∎

Let R be a ring with identity 1.

Definition (Ideal Generated By A Subset Of A Ring). Let A be a subset of R. Let (A) denote the smallest ideal of R containing A. We can define (A) more concretely by

$$(A) = \bigcap_{I \in \Lambda} I$$

where $\Lambda = \{I \text{ is an ideal of } R : A \subseteq I\}$. An ideal generated by a finite set is called a finitely generated ideal.

If $A = \{a_1, a_2, \ldots, a_n\}$, then (A) will be written $(a_1, a_2, \ldots, a_n)$. Also, note that $(I, J) = (I \cup J)$ (if the ideal is generated by two sets I and J).

What is the simplest group generated by a set? In group theory, we call these sets cyclic groups. They generate groups with single elements. In the ring setting, we call these kinds of sets **principal ideals**. That is, $a \neq 0$ in R is a **principal ideal** if (a) is an ideal. Here, (a) must be closed with respect to addition. More precisely,

Definition (Principal Ideals)**.** Let R be a ring and $a \in R$. If R is a commutative ring with $1 \neq 0$, then the set

$$(a) = \{ar : r \in R\}$$

is called the **principal ideal** of R generated by a.

Example.

(1) In any ring, $0 = (0)$ is a principal ideal generating the trivial ring and $(1) = R$ is the principal ideal that generates R.

(2) In $\mathbb{Z}$, all ideals are **principal**. Why? All additive subgroups of $\mathbb{Z}$ are of the form $n\mathbb{Z}$ $(n \geq 0)$ and $n\mathbb{Z}$ is an ideal for all $n \geq 0$. Further, $n\mathbb{Z} = \{nx : x \in \mathbb{Z}\} = (n)$. Because $\mathbb{Z}$ only has principal ideals, we see that

$$(6, 9) = (d) = (3)$$

where $d = \gcd(6, 9) = 3$. One can prove in general that for any $a, b \in \mathbb{Z}$ non-zero, $(a, b) = (d)$ where $d = \gcd(a, b)$.

Proposition. Let I be an ideal of R and let R be a ring.

(1) $I = R$ if and only if I contains a unit.

(2) Assume R is commutative. R is a field if and only if the only ideals are 0 and R.

Proof.

(1) If $I = R$, then $1 \in I$. Conversely, if u is a unit in I with inverse $v \in R$, then for any $r \in R$, we have

$$r = r \cdot 1 = r(vu) = (rv)u \in I.$$

Hence, $R = I$.

(2) The ring R is a field if and only if every non-zero element is a unit. Hence, any non-zero ideal contains a unit. Then the only nonzero ideal of R is R itself.

Conversely, if 0 and R are the only ideals of R, we let $u \neq 0$ in R be arbitrary. By hypothesis, $(u) = R$ and so $1 \in (u)$. That is,

$$1 = uv \text{ for some } v \in R \text{ (also, } vu = 1 \text{ since } R \text{ is commutative)}$$

That is, u is a unit. Thus, every nonzero element of R is a unit so R is a field.

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Corollary. If R is a field, then any nonzero ring homomorphism from R into another ring S is an injection.

Proof. Let R be a field. Let $\varphi : R \to S$ be a non-zero ring homomorphism. Since $\ker\varphi$ is an ideal of the field R, $\ker \varphi = 0$ or $\ker\varphi = R$. If $\ker \varphi = R$, then for all $r \in R$ $\varphi(r) = 0$ contrary to our assumption. Hence, $\ker \varphi = 0$ and so φ must be an injection. ■

Definition (Maximal Ideal)**.** An ideal M in a ring S is called a **maximal ideal** if it is proper (that is, $M \neq S$) and the only ideals which contain M are S and itself; that is, if for all ideals $K \leq R$

with $M \leq K \leq R$, then $K = M$ or $K = R$.

Proposition. In a ring with identity, every proper ideal is contained in a maximal ideal. (Be able to prove this for finite rings in the exam).

Proposition (Proposition 12 from Dummit and Foote)**.** Assume R is commutative. The ideal M is a maximal if and only if R/M is a field.

Proof. We see that M is a maximal ideal if and only if there are no ideals, I, with $M < I < R$. Using the fourth isomorphism theorem, the ideals of R containing M are in $1-1$ correspondence with the ideals of R/M. This true if and only if the only ideals of R/M are R/M itself and $\overline{0}$. This true if and only if R/M is a field. ■

Remark (Things to Verify In The Above Proof)**.**

- Show that if R is commutative, then R/M is commutative.
- R/M has unity, and R/M is not the zero ring. (This is true because $M \neq R$ which implies that $R/M \neq \overline{0}$).

Example. What are the maximal ideals in $\mathbb{Z}$?

We know all subrings of $\mathbb{Z}$ are of the form in $n\mathbb{Z}$ for $n \geq 0$. It is straightforward to show that $n\mathbb{Z}$ is an ideal.

The maximal ideals are those ideals such that $\mathbb{Z}/n\mathbb{Z}$ is a field. We know that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime. That is , $n\mathbb{Z}$ is maximal if and only if p is prime.

Definition (Prime Ideals)**.** Assume R is a commutative ring. An ideal P is a prime ideal if $P \neq R$ (this means that P is not the zero ring) and if $ab \in P$, then either $a \in P$ or $b \in P$.

Proposition (Proposition 13 from Dummit and Foote)**.** Assume R is commutative. The ideal P is a prime ideal in R if and only if R/P is an integral domain.

Proof. The ideal P is prime if and only if $P \neq R$ and if $ab \in P$, then either $a \in P$ or $b \in P$. That is, either $a + P = 0_R + P$ or $b + P = 0_R + P$. Since $P \neq R$, we have $R/P \neq \overline{0}$ and if $(a + P)(b + P) = 0 + P$, then either $a + P = 0_R + P$ or $b + P = 0_R + P$, implying that R/P is an integral domain. ■

Corollary. Every maximal ideal is a prime ideal when our ring commutative.

Suppose that M is an ideal such that R/M is a field, then M is maximal.

4 Week 4